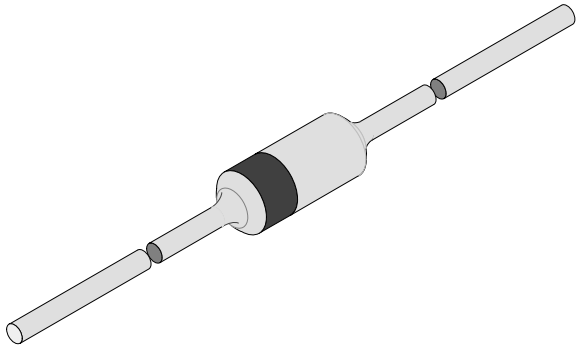
Silicon Epitaxial Planar Z–Diodes

Features

- Very sharp reverse characteristic
- Low reverse current level
- Low noise
- Very high stability
- Available with tighter tolerances
- V_Z–tolerance ± 2%



94 9367

Applications

Voltage stabilization

Absolute Maximum Ratings

T_j = 25°C

Parameter	Test Conditions	Type	Symbol	Value	Unit
Power dissipation	l=4mm, T_L=25°C		P_V	500	mW
Z–current			I_Z	P_V/V_Z	mA
Junction temperature			T_j	175	°C
Storage temperature range			T_{stg}	–65...+175	°C

Maximum Thermal Resistance

T_j = 25°C

Parameter	Test Conditions	Symbol	Value	Unit
Junction ambient	l=4mm, T_L=constant	R_{thJA}	300	K/W

Electrical Characteristics

T_j = 25°C

Parameter	Test Conditions	Type	Symbol	Min	Typ	Max	Unit
Forward voltage	I_F=100mA		V_F			1.5	V

Type	V_{Znom}	I_{ZT} for V_{ZT} and		r_{ziT}	r_{zik} at	I_{ZK}	I_R and I_R at		V_R	TK_{VZ}
BZX55B...	V	mA	V	Ω	Ω	mA	μA	μA [2]	V	%/K
2V7	2.7	5	2.64 to 2.76	< 85	< 600	1	< 10	< 50	1	–0.09 to –0.06
3V0	3.0	5	2.94 to 3.06	< 90	< 600	1	< 4	< 40	1	–0.08 to –0.05
3V3	3.3	5	3.24 to 3.36	< 90	< 600	1	< 2	< 40	1	–0.08 to –0.05
3V6	3.6	5	3.52 to 3.68	< 90	< 600	1	< 2	< 40	1	–0.08 to –0.05
3V9	3.9	5	3.82 to 3.98	< 90	< 600	1	< 2	< 40	1	–0.08 to –0.05
4V3	4.3	5	4.22 to 4.38	< 90	< 600	1	< 1	< 20	1	–0.06 to –0.03
4V7	4.7	5	4.60 to 4.80	< 80	< 600	1	< 0.5	< 10	1	–0.05 to +0.02
5V1	5.1	5	5.00 to 5.20	< 60	< 550	1	< 0.1	< 2	1	–0.02 to +0.02
5V6	5.6	5	5.48 to 5.72	< 40	< 450	1	< 0.1	< 2	1	–0.05 to +0.05
6V2	6.2	5	6.08 to 6.32	< 10	< 200	1	< 0.1	< 2	2	0.03 to 0.06
6V8	6.8	5	6.66 to 6.94	< 8	< 150	1	< 0.1	< 2	3	0.03 to 0.07
7V5	7.5	5	7.35 to 7.65	< 7	< 50	1	< 0.1	< 2	5	0.03 to 0.07
8V2	8.2	5	8.04 to 8.36	< 7	< 50	1	< 0.1	< 2	6.2	0.03 to 0.08
9V1	9.1	5	8.92 to 9.28	< 10	< 50	1	< 0.1	< 2	6.8	0.03 to 0.09
10	10	5	9.80 to 10.20	< 15	< 70	1	< 0.1	< 2	7.5	0.03 to 0.1
11	11	5	10.78 to 11.22	< 20	< 70	1	< 0.1	< 2	8.2	0.03 to 0.11
12	12	5	11.76 to 12.24	< 20	< 90	1	< 0.1	< 2	9.1	0.03 to 0.11
13	13	5	12.74 to 13.26	< 26	< 110	1	< 0.1	< 2	10	0.03 to 0.11
15	15	5	14.70 to 15.30	< 30	< 110	1	< 0.1	< 2	11	0.03 to 0.11
16	16	5	15.70 to 16.30	< 40	< 170	1	< 0.1	< 2	12	0.03 to 0.11
18	18	5	17.64 to 18.36	< 50	< 170	1	< 0.1	< 2	13	0.03 to 0.11
20	20	5	19.60 to 20.40	< 55	< 220	1	< 0.1	< 2	15	0.03 to 0.11
22	22	5	21.55 to 22.45	< 55	< 220	1	< 0.1	< 2	16	0.04 to 0.12
24	24	5	23.5 to 24.5	< 80	< 220	1	< 0.1	< 2	18	0.04 to 0,12
27	27	5	26.4 to 27.6	< 80	< 220	1	< 0.1	< 2	20	0.04 to 0.12
30	30	5	29.4 to 30.6	< 80	< 220	1	< 0.1	< 2	22	0.04 to 0.12
33	33	5	32.4 to 33.6	< 80	< 220	1	< 0.1	< 2	24	0.04 to 0.12
36	36	5	35.3 to 36.7	< 80	< 220	1	< 0.1	< 2	27	0.04 to 0.12
39	39	2.5	38.2 to 39.8	< 90	< 500	0.5	< 0.1	< 5	30	0.04 to 0.12
43	43	2.5	42.1 to 43.9	< 90	< 600	0.5	< 0.1	< 5	33	0.04 to 0.12
47	47	2.5	46.1 to 47.9	< 110	< 700	0.5	< 0.1	< 5	36	0.04 to 0.12
51	51	2.5	50.0 to 52.0	< 125	< 700	0.5	< 0.1	< 10	39	0.04 to 0.12
56	56	2.5	54.9 to 57.1	< 135	< 1000	0.5	< 0.1	< 10	43	0.04 to 0.12
62	62	2.5	60.8 to 63.2	< 150	< 1000	0.5	< 0.1	< 10	47	0.04 to 0.12
68	68	2.5	66.6 to 69.4	< 200	< 1000	0.5	< 0.1	< 10	51	0.04 to 0.12
75	75	2.5	73.5 to 76.5	< 250	< 1500	0.5	< 0.1	< 10	56	0.04 to 0.12

Characteristics (T_j = 25°C unless otherwise specified)

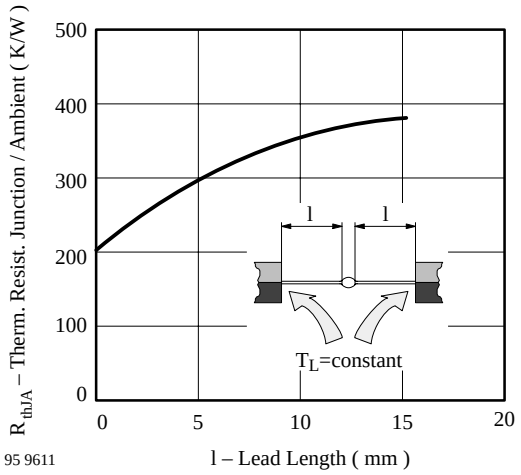


Figure 1. Thermal Resistance vs. Lead Length

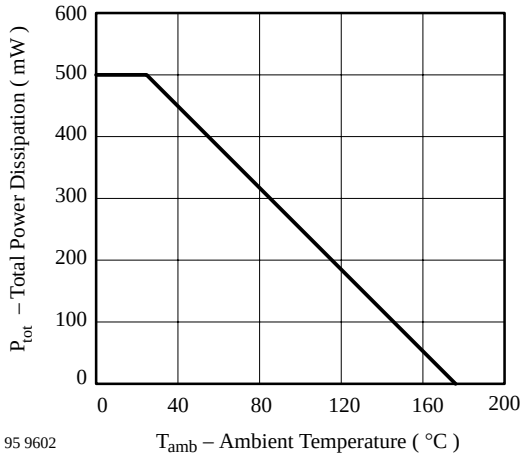


Figure 2. Total Power Dissipation vs. Ambient Temperature

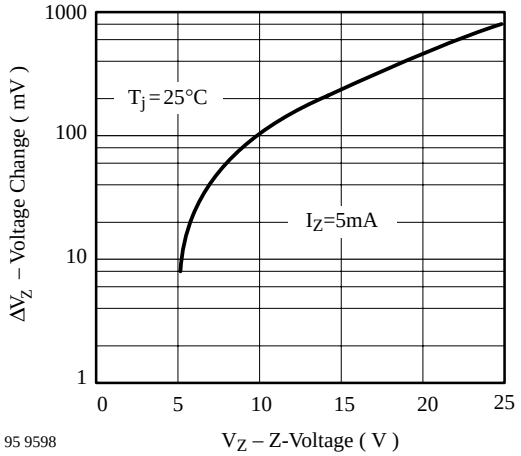


Figure 3. Typical Change of Working Voltage under Operating Conditions at T_{amb}=25˚C

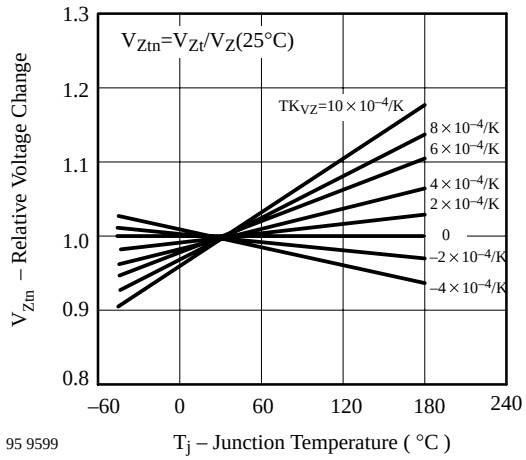


Figure 4. Typical Change of Working Voltage vs. Junction Temperature

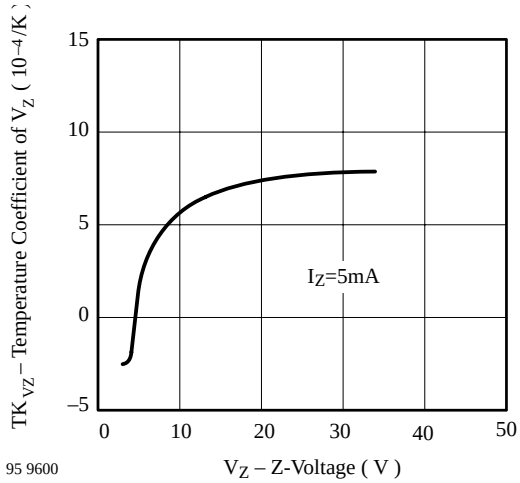


Figure 5. Temperature Coefficient of Vz vs. Z–Voltage

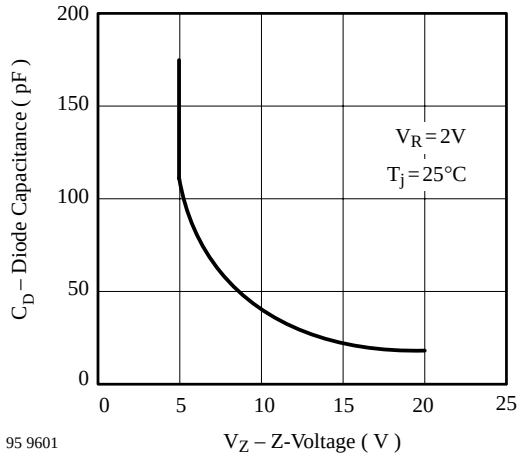


Figure 6. Diode Capacitance vs. Z–Voltage

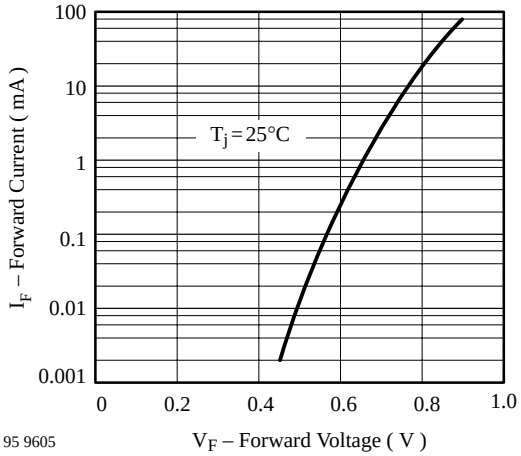


Figure 7. Forward Current vs. Forward Voltage

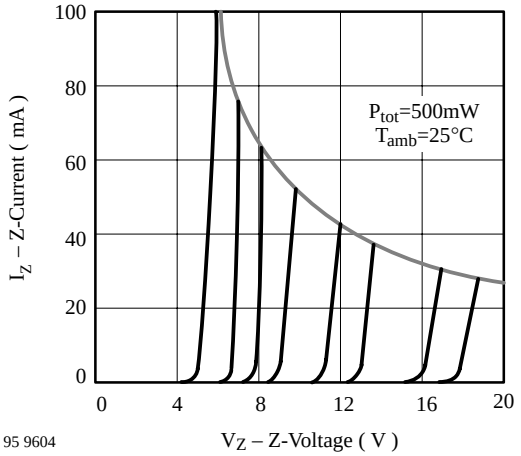


Figure 8. Z–Current vs. Z–Voltage

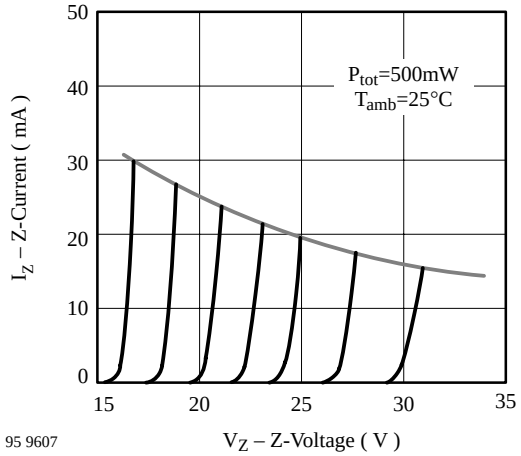


Figure 9. Z–Current vs. Z–Voltage

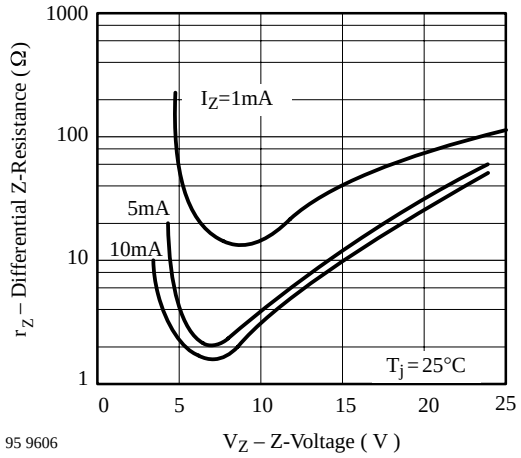


Figure 10. Differential Z–Resistance vs. Z–Voltage

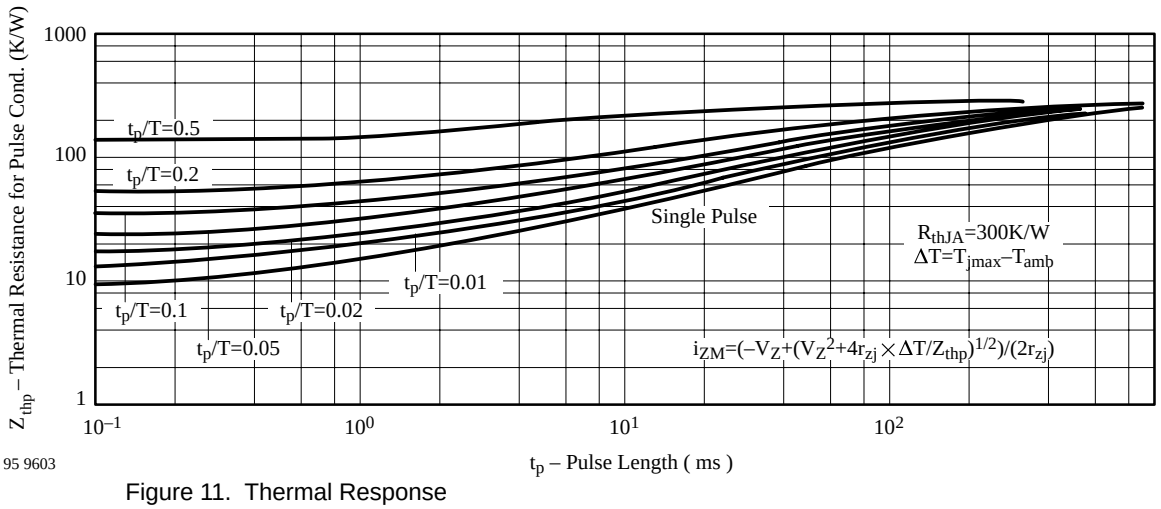


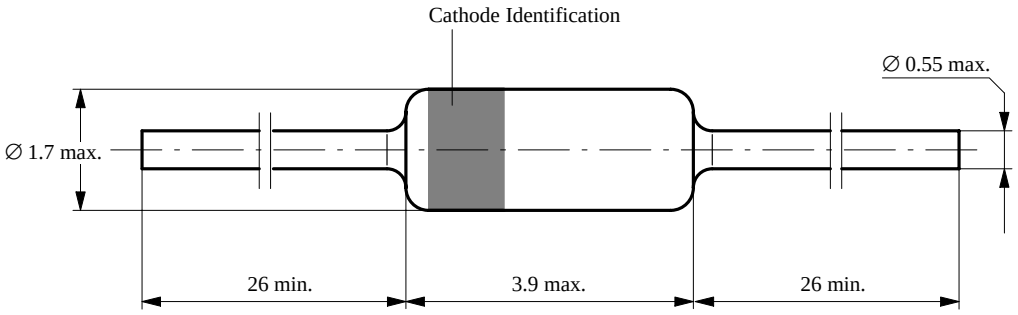
Figure 11. Thermal Response

Dimensions in mm

technical drawings
according to DIN
specifications

94 9366

Standard Glass Case
54 A 2 DIN 41880
JEDEC DO 35
Weight max. 0.3 g



Ozone Depleting Substances Policy Statement

It is the policy of **Vishay Semiconductor GmbH** to

1. Meet all present and future national and international statutory requirements.

2. Regularly and continuously improve the performance of our products, processes, distribution and operating systems with respect to their impact on the health and safety of our employees and the public, as well as their impact on the environment.

It is particular concern to control or eliminate releases of those substances into the atmosphere which are known as ozone depleting substances (ODSs).

The Montreal Protocol (1987) and its London Amendments (1990) intend to severely restrict the use of ODSs and forbid their use within the next ten years. Various national and international initiatives are pressing for an earlier ban on these substances.

Vishay Semiconductor GmbH has been able to use its policy of continuous improvements to eliminate the use of ODSs listed in the following documents.

1. Annex A, B and list of transitional substances of the Montreal Protocol and the London Amendments respectively

2. Class I and II ozone depleting substances in the Clean Air Act Amendments of 1990 by the Environmental Protection Agency (EPA) in the USA

3. Council Decision 88/540/EEC and 91/690/EEC Annex A, B and C (transitional substances) respectively.

Vishay Semiconductor GmbH can certify that our semiconductors are not manufactured with ozone depleting substances and do not contain such substances.

We reserve the right to make changes to improve technical design and may do so without further notice.
Parameters can vary in different applications. All operating parameters must be validated for each customer application by the customer. Should the buyer use Vishay-Telefunken products for any unintended or unauthorized application, the buyer shall indemnify Vishay-Telefunken against all claims, costs, damages, and expenses, arising out of, directly or indirectly, any claim of personal damage, injury or death associated with such unintended or unauthorized use.

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